Flexibility vs. Interpretability in High-Dimensional Cancer Classification

Regularized Regression, Tree Ensembles, and Neural Networks
on TCGA RNA-Seq Data ($p \gg n$)

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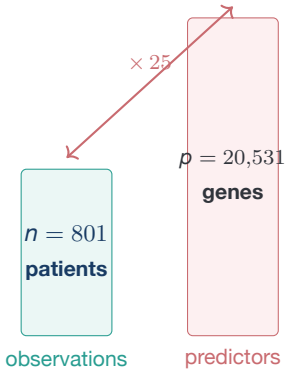
INDIAN STATISTICAL INSTITUTE, KOLKATA

Regression Techniques
Course Project

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Why Is High-Dimensional Gene Expression Hard?

801 patients, 20,531 measured genes — a severely underdetermined system



Three immediate statistical consequences:

- **Singular covariance:** Sample covariance $\mathbf{S} \in \mathbb{R}^{p \times p}$ has rank $\leq 800 \ll 20{,}531$. Classical OLS and Fisher LDA cannot be inverted.
- **Complete linear separability:** In $p \gg n$, data are linearly separable. Standard MLE diverges ($\|\mathbf{w}\|_2 \to \infty$).
- **Strong multicollinearity:** Genes function in shared pathways; pairs show $|r| > 0.85$, destabilising feature selection.

▷ **Regularization is not optional — it is a mathematical necessity.**

SECTION 01

Dataset and Exploratory Analysis

TCGA RNA-Seq structure, variance screening, and inherent class separability

TCGA RNA-Seq Dataset

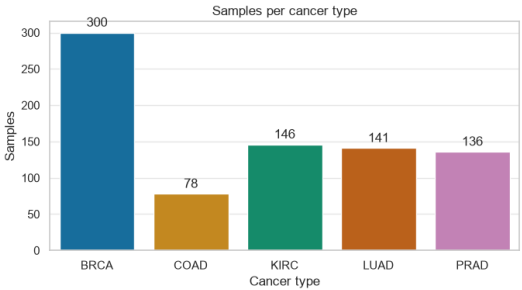
Five primary tumour types — balanced, log-transformed, pre-screened by variance

801 samples
Total patients

5 cancer types
BRCA KIRC COAD
LUAD PRAD

1,000 genes
After MAD pre-screen

20,531 raw genes
Before filtering

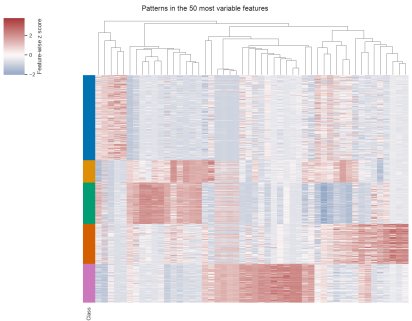


Preprocessing pipeline:

- Log-transform: $z_{ij} = \log_2(x_{ij} + 1)$ to stabilize variance.
- Keep top-1,000 genes by **in-fold MAD** (no label leakage).
- *Z*-score standardize within each training fold.

Expression Heatmap: 50 Highest-Variance Genes

Unsupervised hierarchical clustering reveals tissue-specific expression blocks



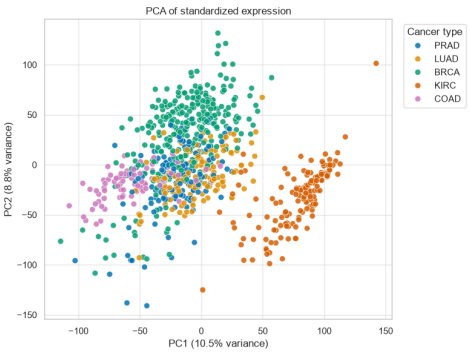
Key observations:

- Colours show standardised expression per gene; columns are patients.
- Cancer cohorts form **distinct blocks** without any labels.
- Row clusters reflect shared **gene modules** and pathways.
- Strong visual separation motivates linear classification.

▷ **The dataset carries strong class-specific structure even in unsupervised clustering.**

PCA: Tumours Separate in Two Linear Dimensions

Spectral projection of 1,000 variance-filtered genes onto the leading principal components



- **PC1 (18.6%):** Separates BRCA from KIRC.
- **PC2 (12.3%):** Separates COAD and LUAD.
- **30.9%** of total variation visible in 2D.
- Five cohorts are almost fully linearly separable in this unsupervised projection.

▷ **Linear separability exists before any model fitting — nonlinear methods gain little here.**

SECTION 02

Statistical Methodology

Regularized multinomial regression, tree ensembles, and neural network baselines

Three Regularized Multinomial Classifiers

Same cross-entropy loss, different penalty geometry — very different behaviour under collinearity

Ridge (L_2)

$$\mathcal{L}_{\text{CE}} + \frac{\lambda}{2} \sum_k \|\mathbf{w}_k\|_2^2$$

- $\lambda \mathbf{I}$ makes Hessian strictly positive definite.
- Shrinks all weights smoothly toward zero.
- Retains **all 1,000 genes**.
- Optimal probability calibration (log-loss 0.005).

Lasso (L_1)

$$\mathcal{L}_{\text{CE}} + \lambda \sum_k \|\mathbf{w}_k\|_1$$

- Polyhedral L_1 ball creates corner solutions — exact zeros.
- Reduces to **89 active genes**.
- **Problem:** arbitrarily picks one gene from each collinear cluster.

Elastic Net ($L_1 + L_2$)

$$\mathcal{L}_{\text{CE}} + \lambda \sum_k \left[\alpha \|\mathbf{w}_k\|_1 + \tfrac{1-\alpha}{2} \|\mathbf{w}_k\|_2^2\right]$$

- Combines sparsity (L_1) with stability (L_2).
- Selects **218 active genes**, $\alpha = 0.5$.
- **Grouping property:** co-regulated genes enter/leave together.

Why Elastic Net Handles Collinearity Better

A formal bound — and the non-linear baselines we compare against

The Grouping Property (Zou & Hastie 2005)

For features with sample correlation r:

$$\left|\hat{w}_i - \hat{w}_j\right| \;\le\; \frac{\sqrt{2(1-r)}}{\lambda(1-\alpha)}\,\|\mathbf{y}\|_2$$

- As $r \to 1$, bound $\to 0$: correlated genes receive equal weights.
- Correlated features enter/leave the active set **together**.
- Resolves Lasso's arbitrary dropping of co-regulated genes.

Non-Linear Baselines

- **Random Forest:** 200 trees, $m = \lfloor\sqrt{p}\rfloor \approx 31$ per split.
- **XGBoost:** gradient boosting, $\eta = 0.10$, max depth 4.
- **PCA-MLP:** 434 PCs (95% var.), $434 \to 128 \to 64 \to 5$ MLP. Operates on PCs, not genes.
- **Sparse NN:** input group-lasso penalty for feature selection.

▷ **Trees choose among collinear genes arbitrarily; Elastic Net assigns them collectively.**

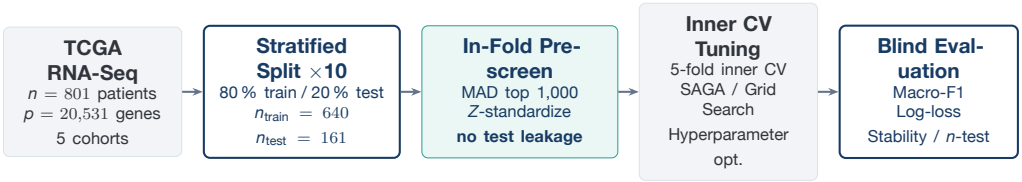
SECTION 03

Experimental Design

Validation protocol, evaluation metrics, and stress tests

Zero-Leakage Evaluation Pipeline

10 repeated stratified splits with strictly within-fold feature prescreening



Strict separation. MAD screening and Z-standardization are computed solely on the training fold. Test data remain entirely blind until final scoring.

Variance estimation. 10 independent random resamplings quantify the variability of every metric, not just its mean.

Why Three Metrics?

Macro-F1 reveals accuracy; log-loss probes calibration; Dice/Jaccard evaluate biomarker stability

Macro-F1

$$\frac{1}{5}\sum_{k=1}^{5}\frac{2P_kR_k}{P_k+R_k}$$

Unweighted mean over all 5 cohorts. Robust to class imbalance (BRCA $n = 300$; COAD $n = 78$).

Limitation: all models reach > 0.993; cannot discriminate calibration.

Log-Loss

$$-\frac{1}{N}\sum_{i,k} y_{ik}\log\hat{p}_{ik}$$

Heavily penalises confident misclassifications. Measures probability calibration, not just top-1 accuracy.

Primary ranking metric across all evaluated models.

Stability

$$D = \frac{2|S_a\cap S_b|}{|S_a|+|S_b|},\ J = \frac{|S_a\cap S_b|}{|S_a\cup S_b|}$$

Pairwise agreement between active gene sets across independent training folds.

Essential for biomarker reproducibility across clinical patient samples.

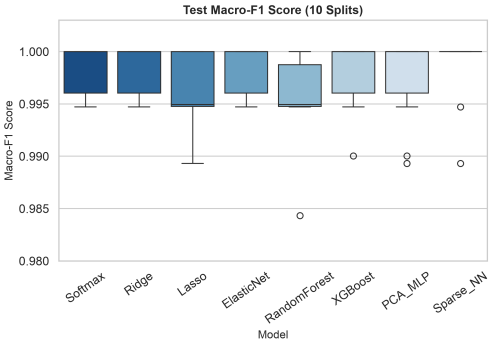
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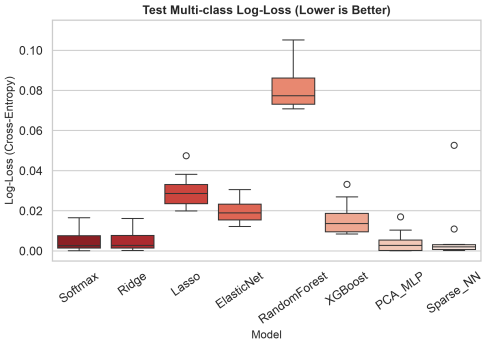
Results

Benchmark accuracy, calibration, sparsity, and stability across all eight models

All Models Achieve High Accuracy – Log-Loss Tells the Real Story

Performance distributions across 10 repeated train/test splits





F1 saturates above 0.993 for every model. Accuracy alone is insufficient to distinguish architectures on this dataset.

Log-loss diverges sharply. Random Forest: 0.081. Linear models and PCA-MLP: < 0.020. The gap is roughly $\times 10$.

Comprehensive Model Comparison

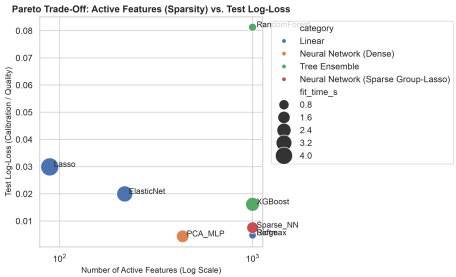
Key metrics: Macro-F1, log-loss, active features, fit time, and selection stability

Model	Macro-F1	Log-Loss	Active Genes	Dice / Jaccard
Softmax (baseline)	0.998 ± 0.003	0.005	1000	1.00 / 1.00
Ridge (L_2)	0.998 ± 0.003	0.005	1000	1.00 / 1.00
Lasso (L_1)	0.995 ± 0.006	0.030	**89**	0.61 / 0.44
Elastic Net	$\mathbf{0.998 \pm 0.003}$	**0.020**	**218**	**0.73 / 0.57**
Random Forest	0.994 ± 0.006	0.081	1000	0.28 / 0.16
XGBoost	0.995 ± 0.006	0.028	1000	0.44 / 0.29
PCA-MLP	0.998 ± 0.003	0.004	434 PCs	—
Sparse NN	0.997 ± 0.004	0.063	1000	—

▷ **Elastic Net matches the best F1, uses only 218 genes, and leads all sparse models in stability.**

Sparsity vs. Calibration: The Pareto Frontier

Fewer active genes means slightly higher log-loss — Elastic Net occupies the efficient frontier

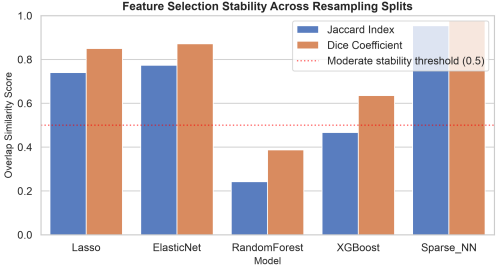


- **Ridge / PCA-MLP:** 1000 genes / 434 PCs, log-loss 0.004–0.005. Dense but calibrated.
- **Lasso:** 89 genes, log-loss 0.030. Highly sparse, slight calibration penalty.
- **Elastic Net:** 218 genes, log-loss 0.020. Sits on optimal frontier.
- **Trees:** dense *and* poorly calibrated (0.081). Off frontier.

▷ **Elastic Net balances biological parsimony and probability calibration.**

Feature Selection Stability

How consistently does each model select the same genes across different data splits?

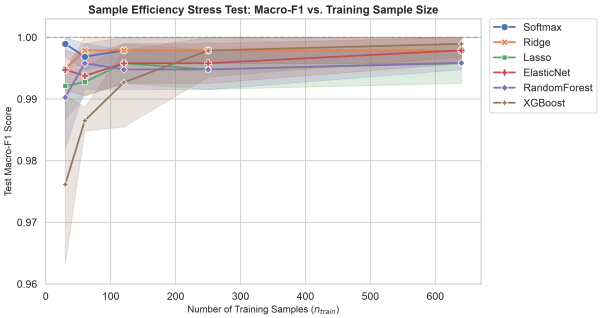


- **Elastic Net:** Dice 0.73, Jaccard 0.57. Highest stability among sparse models.
- **Lasso:** Dice 0.61, Jaccard 0.44. Arbitrary collinear choices hurt stability.
- **XGBoost:** Jaccard 0.29. Greedy splitting is inconsistent.
- **Random Forest:** Jaccard $\mathbf{0.16}$. Splits fluctuate with identical impurity gains.

▷ **Biomarker discovery requires L_2 grouping; trees cannot stably rank collinear genes.**

Sample Efficiency: What Happens with Fewer Patients?

Macro-F1 as training set size drops from 640 to 30 (6 patients per class)



- **Linear models** (Ridge, Elastic Net): maintain ≥ 0.995 F1 even down to $n_{\text{train}} = 30$.
- **XGBoost** drops to 0.976 ± 0.018 (min 0.954). Greedy splits overfit noise under scarcity.
- **Random Forest** drops to 0.985 at $n_{\text{train}} = 30$.

Y-axis starts at 0.96 to highlight low-n divergence.

▷ **Regularized linear models remain sample-efficient; trees degrade sharply.**

SECTION 05

Interpretation and Conclusions

Research question answers, limitations, and practical guidance

Three Main Findings

Q1 **Flexibility does not help in $p \gg n$ transcriptomics.**

RNA-Seq data are linearly separable. Deep networks and boosting achieve no F1 advantage, yet produce worse calibration and lack interpretability.

Q2 **Collinearity destabilises Lasso and breaks tree feature selection.**

Lasso arbitrarily drops co-regulated genes. Random Forest drops to Jaccard 0.16. Elastic Net's grouping property yields the most reproducible biomarkers (0.57).

Q3 **Elastic Net achieves the best overall trade-off.**

Matches top F1 (0.998), uses only 218 genes (78% reduction), improves log-loss over Lasso (0.020 vs 0.030), and leads in stability.

Limitations and Open Questions

What this study does not establish:

- **Single TCGA cohort:** $n = 801$ from one platform; multi-centre batch effects were not assessed.
- **Five coarse cancer types:** inter-tissue separability is large. May not generalise to intra-organ subtyping.
- **No clinical validation:** our 218-gene set is a computational screen, not a diagnostic panel.
- **Gene-level biology:** findings are predictive associations; individual genes need experimental assay validation.

Where non-linear models might matter:

- Subtle intra-organ subtyping (e.g. breast cancer subtypes).
- Higher-order non-linear gene interactions.
- Multi-modal integration (genomics + pathology images).

Practical guidance:

- Evaluate beyond accuracy: log-loss and stability matter.
- Default to Elastic Net for high-dimensional RNA-Seq.
- Avoid tree feature importances under high collinearity.

In $p \gg n$ RNA-Seq classification, regularized linear models are not a compromise – they are the principled choice.

High accuracy, sharp calibration, 78% feature reduction,
and the most reproducible biomarker sets.

Indian Statistical Institute, Kolkata | November 2026

Thank You

Questions & Discussion